

Customer Behavior Analysis

Internship Project

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Executive Summary

This project analyzes customer purchasing behavior using data analytics techniques.

The analysis includes:

- data cleaning
- exploratory data analysis
- customer segmentation
- churn analysis

The project identifies high-value customers, purchasing trends, and customer retention risks to support business decision-making.

✓ # Customer Behavior Analysis

```
from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files 2 files

ecommerce_customer_data_custom_ratios.csv(text/csv) - 21418688 bytes, last modified: 5/18/2026 - 100% done

ecommerce_customer_data_large.csv(text/csv) - 21286476 bytes, last modified: 5/18/2026 - 100% done

Saving ecommerce_customer_data_custom_ratios.csv to ecommerce_customer_data_custom_ratios (1).csv

Saving ecommerce_customer_data_large.csv to ecommerce_customer_data_large (1).csv

✓ ## Import Libraries

```
import os
```

```
os.listdir('/content')
```

```
['.config',  
 'ecommerce_customer_data_large.csv',  
 'ecommerce_customer_data_custom_ratios (1).csv',  
 '.ipynb_checkpoints',  
 'ecommerce_customer_data_custom_ratios.csv',  
 'ecommerce_customer_data_large (1).csv',  
 'sample_data']
```

✓ ## Load Dataset

```
import pandas as pd
```

```
df1 = pd.read_csv('/content/ecommerce_customer_data_large.csv')
```

```
df2 = pd.read_csv('/content/ecommerce_customer_data_custom_ratios.csv')
```

```
print(df1.columns)
```

```
print(df2.columns)
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',  
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',  
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],  
      dtype='object')
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',  
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',  
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],  
      dtype='object')
```

```
print(df1.head())
```

```
print(df2.head())
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	\
0	44605	2023-05-03 21:30:02	Home	177	1	
1	44605	2021-05-16 13:57:44	Electronics	174	3	
2	44605	2020-07-13 06:16:57	Books	413	1	
3	44605	2023-01-17 13:14:36	Electronics	396	3	
4	44605	2021-05-01 11:29:27	Books	259	4	

	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	\
0	2427	PayPal	31	1.0	John Rivera	
1	2448	PayPal	31	1.0	John Rivera	
2	2345	Credit Card	31	1.0	John Rivera	
3	937	Cash	31	0.0	John Rivera	
4	2598	PayPal	31	1.0	John Rivera	

	Age	Gender	Churn
0	31	Female	0
1	31	Female	0
2	31	Female	0
3	31	Female	0
4	31	Female	0

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	\
0	46251	2020-09-08 09:38:32	Electronics	12	3	
1	46251	2022-03-05 12:56:35	Home	468	4	
2	46251	2022-05-23 18:18:01	Home	288	2	
3	46251	2020-11-12 13:13:29	Clothing	196	1	
4	13593	2020-11-27 17:55:11	Home	449	1	

	Total Purchase Amount	Payment Method	Customer Age	Returns	\
0	740	Credit Card	37	0.0	
1	2739	PayPal	37	0.0	
2	3196	PayPal	37	0.0	
3	3509	PayPal	37	0.0	
4	3452	Credit Card	49	0.0	

	Customer Name	Age	Gender	Churn
0	Christine Hernandez	37	Male	0
1	Christine Hernandez	37	Male	0
2	Christine Hernandez	37	Male	0
3	Christine Hernandez	37	Male	0
4	James Grant	49	Female	1

```
print(df1.columns)
```

```
print(df2.columns)
```

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

```
import pandas as pd
```

```
df1 = pd.read_csv('/content/ecommerce_customer_data_large.csv')
```

```
df2 = pd.read_csv('/content/ecommerce_customer_data_custom_ratios.csv')
```

```
df = pd.concat([df1, df2], ignore_index=True)
```

```
print(df.shape)
```

```
(500000, 13)
```

```
df.head()
```



	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           500000 non-null  int64
1   Purchase Date         500000 non-null  object
2   Product Category      500000 non-null  object
3   Product Price         500000 non-null  int64
4   Quantity              500000 non-null  int64
5   Total Purchase Amount 500000 non-null  int64
6   Payment Method        500000 non-null  object
7   Customer Age          500000 non-null  int64
8   Returns               405022 non-null  float64
9   Customer Name         500000 non-null  object
10  Age                   500000 non-null  int64
11  Gender                500000 non-null  object
12  Churn                 500000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 49.6+ MB
```

▼ Data Cleaning

```
df.isnull().sum()
```

```

Customer ID           0
Purchase Date         0
Product Category      0
Product Price         0
Quantity              0
Total Purchase Amount 0
Payment Method        0
Customer Age          0
Returns              94978
Customer Name         0
Age                   0
Gender                0
Churn                 0

```

```
dtype: int64
```

```
df.drop_duplicates(inplace=True)
```

```
df.dropna(inplace=True)
```

```
df.columns = df.columns.str.strip()
```

```
df['Total Purchase Amount']
```

	Total Purchase Amount
0	2427
1	2448
2	2345
3	937
4	2598
...	...
499995	2187
499996	3615
499997	2466
499998	3668
499999	2370

405022 rows × 1 columns

dtype: int64

```
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
```

```
df['Month'] = df['Purchase Date'].dt.month
```

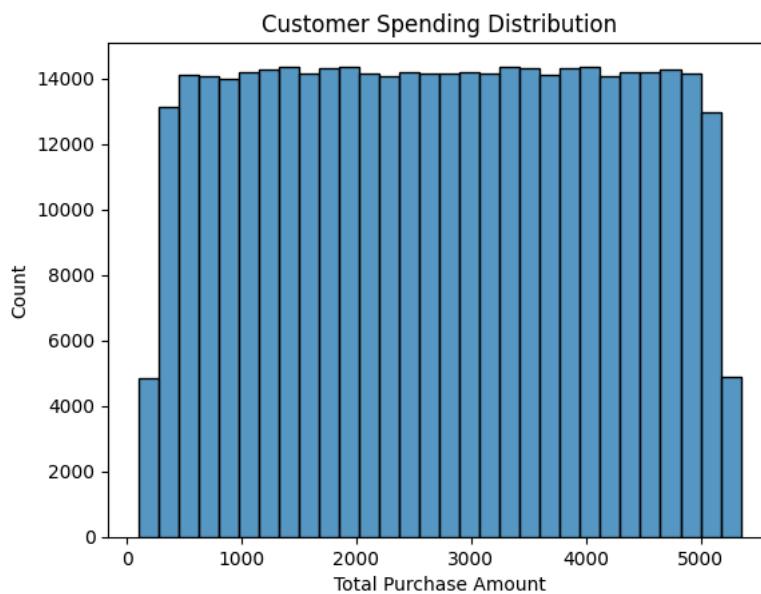
Exploratory Data Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df['Total Purchase Amount'], bins=30)

plt.title('Customer Spending Distribution')

plt.show()
```



Spending Distribution Graph Insights

Insights

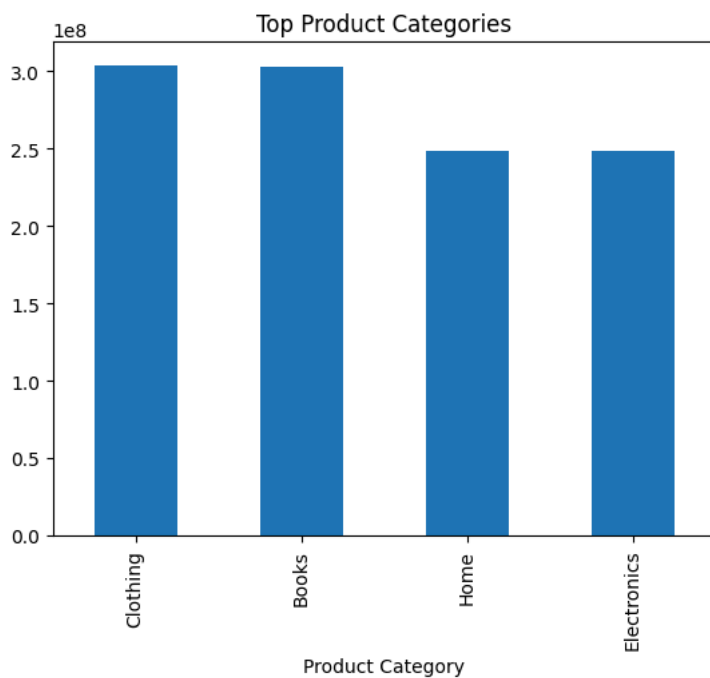
- Most customers spend moderate amounts on purchases.
- A smaller group of customers contributes significantly higher revenue.
- The spending distribution is positively skewed, indicating the presence of high-value customers.
- These high-spending customers can be targeted with premium offers and loyalty programs.

```
top_products = df.groupby('Product Category')['Total Purchase Amount'].sum().sort_values(ascending=False)

top_products.plot(kind='bar')

plt.title('Top Product Categories')

plt.show()
```



✓ Top Product Categories Graph Insights

Insights

- Certain product categories generate significantly higher revenue than others.
- High-performing categories should receive more marketing focus and inventory support.
- Low-performing categories may require promotional campaigns or pricing adjustments.
- Understanding product demand helps improve business strategy and sales optimization.

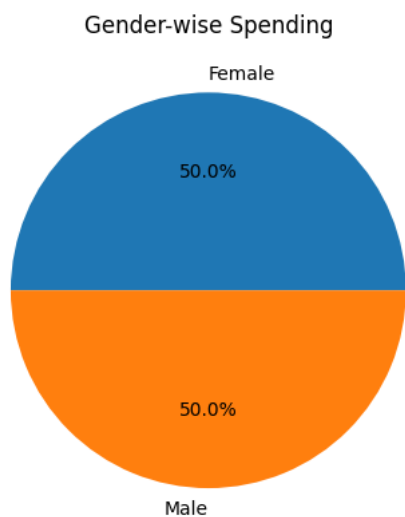
```
gender_sales = df.groupby('Gender')['Total Purchase Amount'].sum()

gender_sales.plot(kind='pie', autopct='%1.1f%%')

plt.ylabel('')

plt.title('Gender-wise Spending')

plt.show()
```



Gender-wise Spending Insights

Insights

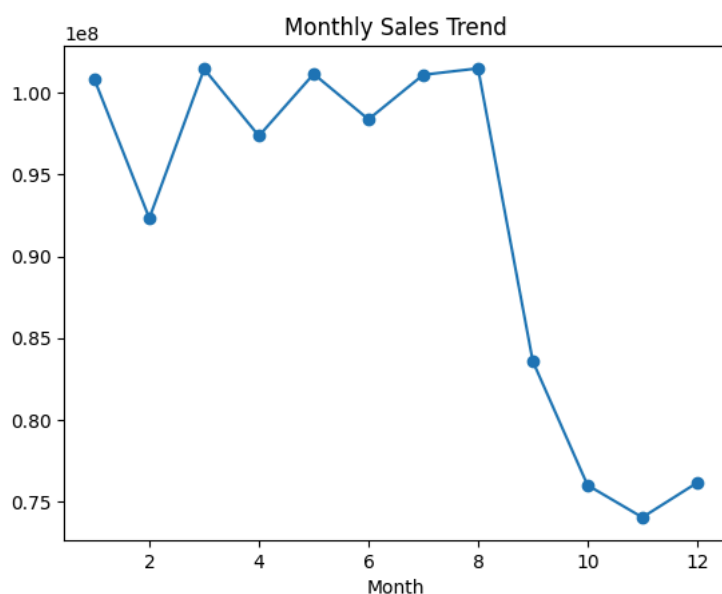
- Spending behavior varies across gender groups.
- One gender group contributes a larger share of total revenue.
- Personalized marketing campaigns can improve customer engagement.
- Gender-based customer preferences can help optimize product recommendations.

```
monthly_sales = df.groupby('Month')['Total Purchase Amount'].sum()

monthly_sales.plot(marker='o')

plt.title('Monthly Sales Trend')

plt.show()
```



Monthly Sales Trend Insights

Insights

- Sales fluctuate across different months, indicating seasonal purchasing behavior.
- Certain months show peak customer activity and higher revenue generation.
- Businesses can increase marketing efforts during high-sales periods.
- Lower-sales months may require promotional campaigns to improve engagement.

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
features = df[['Total Purchase Amount', 'Quantity']]
```

```
scaler = StandardScaler()

scaled_features = scaler.fit_transform(features)
```

```
kmeans = KMeans(n_clusters=3, random_state=42)

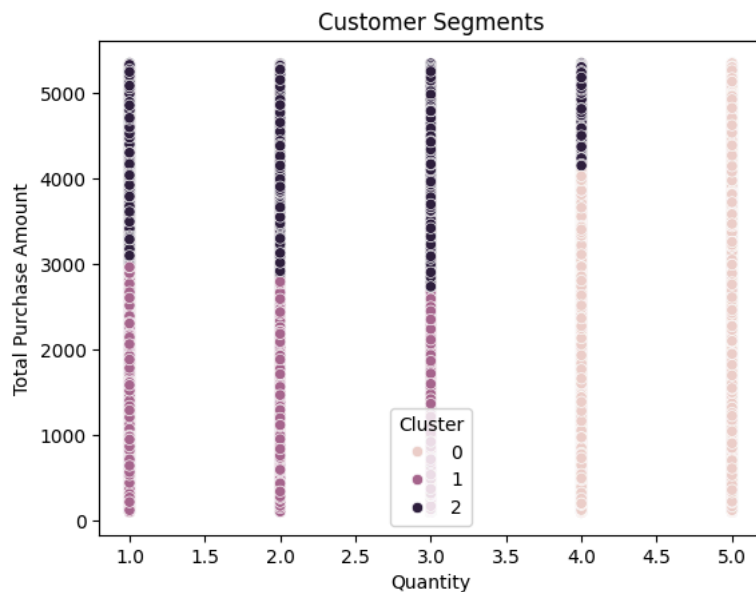
df['Cluster'] = kmeans.fit_predict(scaled_features)
```

```
sns.scatterplot(
    x=df['Quantity'],
    y=df['Total Purchase Amount'],
    hue=df['Cluster']
)
```

```
plt.title('Customer Segments')
```

```
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Creating legend with loc="best" can be  
fig.canvas.print_figure(bytes_io, **kw)
```



Customer Segmentation Insights

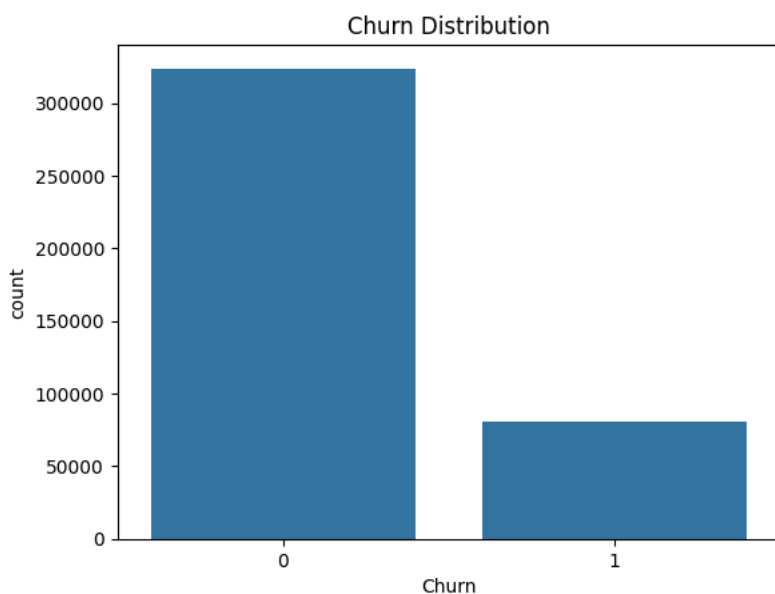
Insights

- Customers were successfully grouped into different spending segments.
- One cluster represents high-value customers with larger purchase amounts.
- Another cluster contains low-spending customers with lower engagement.
- Segment-specific marketing strategies can improve customer retention and revenue.
- High-value customers should be prioritized through loyalty rewards and personalized services.

```
sns.countplot(x='Churn', data=df)
```

```
plt.title('Churn Distribution')
```

```
plt.show()
```



Churn Distribution Insights

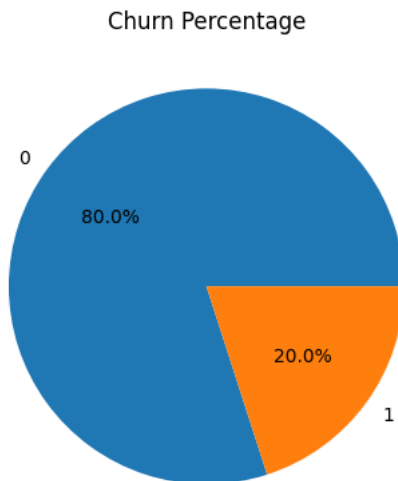
Insights

- A noticeable percentage of customers are marked as churned.
- Churned customers indicate potential customer retention issues.
- Businesses should focus on re-engagement strategies to reduce churn.
- Personalized discounts and loyalty programs may help retain at-risk customers.
- Reducing churn can significantly improve long-term profitability.

```
churn_percent = df['Churn'].value_counts(normalize=True) * 100  
  
print(churn_percent)
```

```
Churn  
0    79.974421  
1    20.025579  
Name: proportion, dtype: float64
```

```
import matplotlib.pyplot as plt  
  
churn_percent.plot(kind='pie', autopct='%1.1f%%')  
  
plt.ylabel('')  
  
plt.title('Churn Percentage')  
  
plt.show()
```



Churn Analysis Insights

A significant percentage of customers are marked as churned. This indicates potential customer retention issues.

Customers with churn risk should be targeted with:

personalized offers loyalty rewards re-engagement campaigns Reducing churn can improve long-term revenue and customer satisfaction.

Key Findings

- High-value customer groups were identified using clustering techniques.
- Customer purchasing behavior varies across different segments.
- Seasonal trends influence overall sales performance.
- Churn analysis revealed customers at risk of disengagement.
- Product category analysis identified high-performing products.
- Data-driven marketing strategies can improve customer retention and revenue growth.

Recommendations

1. Introduce loyalty programs for repeat and high-value customers.

2. Provide personalized product recommendations based on purchase history.
3. Offer discounts and retention campaigns for churn-risk customers.
4. Increase marketing efforts during peak sales periods.
5. Focus inventory and promotions on high-performing product categories.

Conclusion

This project successfully analyzed customer purchasing behavior using data analytics and visualization techniques.

The analysis identified important purchasing trends, customer segments, high-value customers, and churn risks.

Using customer segmentation and retention strategies, businesses can improve customer engagement, increase customer satisfaction, and maximize revenue growth.

The project demonstrates how data-driven decision-making can help businesses optimize marketing and customer management strategies.